Materials : Metals and Non-Metals

Metals

- **Physical properties**

- Shining surface (in pure state) [called metallic lustre]
- Generally hard [varies from metal to metal]
- Malleable [i.e. can be made thin sheets by beating]
- Ductile [i.e. can be drawn into thin wires]

  - [Gold → Highly ductile]

- Good conductors of heat
- High melting point
- Conduct electricity
- Produce sound [some metals; these are called sonorous]

Non-metals

- Non-metals are found in all the three states i.e. solid, liquid and gas, at room temperature.
- Iodine (non-metal) has lustre
- Carbon has allotropes (exists in different forms)
  - Diamond is hard
  - Graphite (Conducts electricity)

Metals	Non-metals
Generally, these are hard and lustrous.	These are soft and have no lustre.
These are malleable and ductile (Malleable: can be beaten into sheets; Ductile: can be drawn into wires).	These are non-malleable and non-ductile.
These are sonorous (produce ringing sound when struck).	These are not sonorous.
These are good conductors of heat and electricity.	These are poor conductors of heat and electricity.

- **Chemical properties:**

Metals	Non-metals

These react with oxygen to produce metal oxides, which are basic in nature.	These react with oxygen to form non-metallic oxides, which are acidic in nature.

Chemical properties

- **Reaction with oxygen**
- **Combine with oxygen to form oxides**

- $2Cu + O_2 \rightarrow 2CuO$
- $4Al + 3O_2 \rightarrow 2Al_2O_3$

- Most metal oxides are insoluble in water.
- If soluble, they form alkali.
- $Na_2O + H_2O \rightarrow 2NaOH$

  $K_2O + H_2O \rightarrow 2KOH$

- Sodium, potassium react very easily with O_2. So, they are kept immersed in kerosene.
- Mg, Al, Zn, Pb form thin layers of oxides.
- **Reaction with water**
- Produce metal oxide + H_2
- If oxide is soluble, then metal hydroxide is formed.

$$\left.\begin{array}{l} 2K + 2H_2O \longrightarrow 2KOH + H_2 \\ 2Na + 2H_2O \longrightarrow 2NaOH + H_2 \end{array}\right\} \begin{array}{l}\text{That's why they are not put in water} \\ \text{Violent reactions}\end{array}$$

$$Ca + 2H_2O \longrightarrow Ca(OH)_2 + H_2 \quad \text{(Less violent)}$$

Mg → Doesn't react with cold H_2O

- Al, Zn, Fe do not react with H_2O, but react with steam.

  $2Al + 3H_2O \rightarrow Al_2O_3 + 3H_2$

  $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$

- **Chemical properties:**

Metals	**Non-metals**
These react with acids to produce metal salts and hydrogen gas.	These do not react with acids.
Some metals react with bases to produce hydrogen gas.	Reactions of non-metals with bases are complex.

Reaction with Acids

- Metal + Dilute acid → Metal salt + H_2
- H_2 doesn't evolve in the case of HNO_3 as it is a strong oxidising agent. It oxidises H_2.
- Cu does not react with acids like dilute H_2SO_4 and dilute HCl.

- **Aqua regia**
  - Freshly-prepared concentrated HCl and concentrated HNO_3 in 3:1 ratio
  - It can dissolve gold and platinum.

Reaction with Bases

- Metals react with bases to produce hydrogen gas.
- Reactions of non-metals with bases are complex.

- **Displacement reactions:**
  - In displacement reactions, a more reactive metal can displace a less reactive metal from their compounds in aqueous solutions. (However, a less reactive metal cannot displace a more reactive metal.)

Example: CuSO4 + Zn → ZnSO4 + Cu Copper Sulphate Zinc Zinc Sulphate Copper (Blue) (Colourless) (Red) CuSO4 + Zn → ZnSO4 + Cu Copper Sulphate Zinc Zinc Sulphate Copper (Blue) (Colourless) (Red)
Fe (s) + CuSO4 (aq) → Cu (s) + FeSO4 (aq)Iron Copper sulphate Copper Iron sulphateFe s + CuSO4 aq → Cu s + FeSO4 aqIron Copper sulphate Copper Iron sulphate

- **Double displacement reaction**
  - Exchange of ions occurs between two compounds.

  - Example
Na2SO4 (aq) + BaCl2 (s) → BaSO4 (aq) + 2NaCl (s)Sodium sulphate Barium Barium sulphate Sodium chloride Na2SO4 aq + BaCl2 s → BaSO4 aq + 2NaCl sSodium sul phate Barium chloride Barium sulphate Sodium chloride

- When the aqueous solution of two compounds react by exchanging their respective ions, such that one of the products formed is insoluble salt and appears in the form of a precipitate, then the reaction is said to be **precipitation reaction**.
- When an acid solution reacts with a base and the two exchange their respective ions, such that only salt and water are products, then the reaction is called **neutralisation reaction.**

- When two compounds react with each other and displace their ions, in such a manner that one of the product formed either decomposes into gaseous compounds or is formed in gaseous state, then the reaction is called **gas-forming reaction**.

- **Uses of metals:**
  - In making machinery, automobiles, jewellery, trains, aeroplanes, cooking utensils, etc.
  - Gold is used for making jewellery, wires, and coins and in dentistry.
  - Silver is used for making coins, ornaments, very thin wires, table cutlery and in photographic films.
  - Copper is used for making wires, utensils, statues, alloys and coins.
  - Iron is used for construction of ships, buildings, automobiles and railway bridges etc.
  - Tin is used for tinning food cans, and making alloys.
  - Lead is used for making batteries, and alloys.
  - Zinc is used in prevention of rusting, making brass and bronze and in dry cells.
  - Aluminium is used in making wires, foils, and alloys.
  - Mercury is used for making amalgams and in thermometers.
  - Magnesium is used for making fire works, and alloys.
- **Uses of non-metals:**
  - They are used in fertilizers, in water purification process, crackers, etc. Oxygen, a non-metal, is essential for our life as all living beings inhale it during breathing.
  - Nitrogen dilutes the activity of oxygen in air. It is used by plants to manufacture proteins.
  - Oxygen is essential for respiration and combustion of fuels.
  - Chlorine is used for bleaching fabrics, sterilization of drinking water, and in manufacturing insecticides and pesticides.
  - Iodine is essential for proper functioning of human body, and in photographic films.
  - Graphite is used as pencil lead, dry lubricant, in electrolytic cells and nuclear reactors.
  - Helium is a noble gas which is used in weather observation balloons.
  - Argon is a noble gas which is used for filling electric bulbs.

Alloys

- Alloys are homogeneous mixtures of two or more metals (or metal and non-metal). Alloying is done to enhance the properties of metals.

- Alloys are made to alter the properties of metals for achieving a specific objective.

- Alloys of aluminium are also useful as they are both light and strong. Some of its alloys are duralumin, magnelium, etc.

- Some alloys of iron are steel, stainless steel, etc. Steel is an alloy of iron and carbon.

- Some alloys of zinc are brass, bronze, and German silver.

- On the basis of composition, alloys are of two types: Substitutional alloys in which atoms of one element randomly replace the atoms of another metal. And interstitial alloys in which small atoms like hydrogen, boron, carbon and nitrogen occupy the holes in the crystal structure of the metal.

- On the basis of constituent elements, alloys are of two types: Ferrous alloys which contain iron as base metal. For example: steel, alnico etc. And non ferrous alloys which do not contain iron as base metal. For example: brass, bronze, duralumin etc.